

arbol221125

November 22, 2025

```
[1]: # Arbol de decisiones, vamos a predecir si una canción llega al top 1 de Billboard o no.
```

```
[2]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import pydot
from sklearn import tree
from sklearn.metrics import accuracy_score
from sklearn.model_selection import KFold
from sklearn.model_selection import cross_val_score
from IPython.display import Image as PImage
from subprocess import check_call
from PIL import Image, ImageDraw, ImageFont
```

```
[3]: # Cargar los datos
artists_billboard = pd.read_csv("artists_billboard_fix3.csv")
```

```
[4]: #Análisis exploratorio de los datos
artists_billboard.head()
```

```
[4]:      id      title \
0    0  Small Town Throwdown
1    1           Bang Bang
2    2           Timber
3    3  Sweater Weather
4    4       Automatic
```

```
      artist      mood \
0  BRANTLEY GILBERT featuring JUSTIN MOORE & THOM...  Brooding
1           JESSIE J, ARIANA GRANDE & NICKI MINAJ  Energizing
2           PITBULL featuring KESHA                Excited
3           THE NEIGHBOURHOOD                     Brooding
4           MIRANDA LAMBERT                       Yearning
```

```
      tempo      genre artist_type chart_date durationSeg top \
0  Medium Tempo  Traditional      Male   20140628       191.0    0
1  Medium Tempo      Pop        Female   20140816       368.0    0
```

|   |              |                    |        |          |       |   |
|---|--------------|--------------------|--------|----------|-------|---|
| 2 | Medium Tempo | Urban              | Mixed  | 20140118 | 223.0 | 1 |
| 3 | Medium Tempo | Alternative & Punk | Male   | 20140104 | 206.0 | 0 |
| 4 | Medium Tempo | Traditional        | Female | 20140301 | 232.0 | 0 |

|   | anioNacimiento |
|---|----------------|
| 0 | 1975.0         |
| 1 | 1989.0         |
| 2 | 1993.0         |
| 3 | 1989.0         |
| 4 | 0.0            |

```
[5]: artists_billboard.shape
```

```
[5]: (635, 11)
```

```
[6]: #Cuántas canciones tengo de cada clase
artists_billboard.groupby('top').size()
```

```
[6]: top
0    494
1    141
dtype: int64
```

```
[7]: # Preparar los datos

# Función para calcular las edades en las que estuvieron en el billboard
def calcula_edad(anio, cuando):
    cad = str(cuando)
    momento = cad[:4]
    if anio == 0.0:
        return None
    return int(momento) - anio
```

```
[8]: artists_billboard['edad_en_billboard'] = artists_billboard.apply(lambda x:
    ↪calcula_edad(x['anioNacimiento'],x['chart_date']),axis=1)
```

```
[9]: artists_billboard.head()
```

```
[9]:   id      title \
0    0  Small Town Throwdown
1    1          Bang Bang
2    2          Timber
3    3  Sweater Weather
4    4          Automatic
```

|   | artist                                            | mood \     |
|---|---------------------------------------------------|------------|
| 0 | BRANTLEY GILBERT featuring JUSTIN MOORE & THOM... | Brooding   |
| 1 | JESSIE J, ARIANA GRANDE & NICKI MINAJ             | Energizing |

|   |                          |          |
|---|--------------------------|----------|
| 2 | PITBULL featuring KE\$HA | Excited  |
| 3 | THE NEIGHBOURHOOD        | Brooding |
| 4 | MIRANDA LAMBERT          | Yearning |

|   | tempo        | genre              | artist_type | chart_date | durationSeg | top | \ |
|---|--------------|--------------------|-------------|------------|-------------|-----|---|
| 0 | Medium Tempo | Traditional        | Male        | 20140628   | 191.0       | 0   |   |
| 1 | Medium Tempo | Pop                | Female      | 20140816   | 368.0       | 0   |   |
| 2 | Medium Tempo | Urban              | Mixed       | 20140118   | 223.0       | 1   |   |
| 3 | Medium Tempo | Alternative & Punk | Male        | 20140104   | 206.0       | 0   |   |
| 4 | Medium Tempo | Traditional        | Female      | 20140301   | 232.0       | 0   |   |

|   | anioNacimiento | edad_en_billboard |
|---|----------------|-------------------|
| 0 | 1975.0         | 39.0              |
| 1 | 1989.0         | 25.0              |
| 2 | 1993.0         | 21.0              |
| 3 | 1989.0         | 25.0              |
| 4 | 0.0            | NaN               |

```
[10]: age_avg = artists_billboard['edad_en_billboard'].mean()
      age_std = artists_billboard['edad_en_billboard'].std()
```

```
[11]: age_avg
```

```
[11]: 30.10282258064516
```

```
[13]: age_std
```

```
[13]: 8.40078832861513
```

```
[14]: # Asignar valores al azar en el rango media-std - media+std que es de 21 a 37
      age_null_count = artists_billboard['edad_en_billboard'].isnull().sum()
      age_null_random_list = np.random.randint(age_avg - age_std, age_avg + age_std,
      ↪size=age_null_count)
```

```
[15]: artists_billboard.head()
```

```
[15]:   id      title \
0    0  Small Town Throwdown
1    1      Bang Bang
2    2      Timber
3    3  Sweater Weather
4    4      Automatic
```

|   | artist                                            | mood       | \ |
|---|---------------------------------------------------|------------|---|
| 0 | BRANTLEY GILBERT featuring JUSTIN MOORE & THOM... | Brooding   |   |
| 1 | JESSIE J, ARIANA GRANDE & NICKI MINAJ             | Energizing |   |
| 2 | PITBULL featuring KE\$HA                          | Excited    |   |
| 3 | THE NEIGHBOURHOOD                                 | Brooding   |   |

4 MIRANDA LAMBERT Yearning

|   | tempo        | genre              | artist_type | chart_date | durationSeg | top | \ |
|---|--------------|--------------------|-------------|------------|-------------|-----|---|
| 0 | Medium Tempo | Traditional        | Male        | 20140628   | 191.0       | 0   |   |
| 1 | Medium Tempo | Pop                | Female      | 20140816   | 368.0       | 0   |   |
| 2 | Medium Tempo | Urban              | Mixed       | 20140118   | 223.0       | 1   |   |
| 3 | Medium Tempo | Alternative & Punk | Male        | 20140104   | 206.0       | 0   |   |
| 4 | Medium Tempo | Traditional        | Female      | 20140301   | 232.0       | 0   |   |

|   | anioNacimiento | edad_en_billboard |
|---|----------------|-------------------|
| 0 | 1975.0         | 39.0              |
| 1 | 1989.0         | 25.0              |
| 2 | 1993.0         | 21.0              |
| 3 | 1989.0         | 25.0              |
| 4 | 0.0            | NaN               |

```
[16]: conValoresNulos = np.isnan(artists_billboard['edad_en_billboard'])
artists_billboard.loc[np.isnan(artists_billboard['edad_en_billboard']),
↳ 'edad_en_billboard'] = age_null_random_list
artists_billboard['edad_en_billboard'] = artists_billboard['edad_en_billboard'].
↳ astype(int)
```

```
[17]: artists_billboard.head()
```

```
[17]: id title \
0 0 Small Town Throwdown
1 1 Bang Bang
2 2 Timber
3 3 Sweater Weather
4 4 Automatic
```

|   | artist                                            | mood       | \ |
|---|---------------------------------------------------|------------|---|
| 0 | BRANTLEY GILBERT featuring JUSTIN MOORE & THOM... | Brooding   |   |
| 1 | JESSIE J, ARIANA GRANDE & NICKI MINAJ             | Energizing |   |
| 2 | PITBULL featuring KESHA                           | Excited    |   |
| 3 | THE NEIGHBOURHOOD                                 | Brooding   |   |
| 4 | MIRANDA LAMBERT                                   | Yearning   |   |

|   | tempo        | genre              | artist_type | chart_date | durationSeg | top | \ |
|---|--------------|--------------------|-------------|------------|-------------|-----|---|
| 0 | Medium Tempo | Traditional        | Male        | 20140628   | 191.0       | 0   |   |
| 1 | Medium Tempo | Pop                | Female      | 20140816   | 368.0       | 0   |   |
| 2 | Medium Tempo | Urban              | Mixed       | 20140118   | 223.0       | 1   |   |
| 3 | Medium Tempo | Alternative & Punk | Male        | 20140104   | 206.0       | 0   |   |
| 4 | Medium Tempo | Traditional        | Female      | 20140301   | 232.0       | 0   |   |

|   | anioNacimiento | edad_en_billboard |
|---|----------------|-------------------|
| 0 | 1975.0         | 39                |

|   |        |    |
|---|--------|----|
| 1 | 1989.0 | 25 |
| 2 | 1993.0 | 21 |
| 3 | 1989.0 | 25 |
| 4 | 0.0    | 35 |

```
[18]: #Voy a empezara mapear el mood para que sean variables categóricas
artists_billboard['moodEncoded'] = artists_billboard['mood'].map(
    {
        'Energizing': 6,
        'Empowering': 6,
        'Cool': 5,
        'Yearning': 4,
        'Excited': 5,
        'Defiant': 3,
        'Sensual': 2,
        'Gritty': 3,
        'Sophisticated': 4,
        'Aggressive': 4,
        'Fiery': 4,
        'Urgent': 3,
        'Rowdy': 4,
        'Sentimental': 4,
        'Easygoing': 1,
        'Melancholy': 4,
        'Romantic': 2,
        'Peaceful': 1,
        'Brooding': 4,
        'Upbeat': 5,
        'Stirring': 5,
        'Lively': 5,
        'Other': 0,
        '': 0
    }).astype(int)
```

```
[19]: artists_billboard.head()
```

```
[19]:   id          title \
0    0  Small Town Throwdown
1    1          Bang Bang
2    2          Timber
3    3  Sweater Weather
4    4          Automatic
```

|   | artist                                            | mood \     |
|---|---------------------------------------------------|------------|
| 0 | BRANTLEY GILBERT featuring JUSTIN MOORE & THOM... | Brooding   |
| 1 | JESSIE J, ARIANA GRANDE & NICKI MINAJ             | Energizing |
| 2 | PITBULL featuring KE\$HA                          | Excited    |

|   |                   |          |
|---|-------------------|----------|
| 3 | THE NEIGHBOURHOOD | Brooding |
| 4 | MIRANDA LAMBERT   | Yearning |

|   | tempo        | genre              | artist_type | chart_date | durationSeg | top | \ |
|---|--------------|--------------------|-------------|------------|-------------|-----|---|
| 0 | Medium Tempo | Traditional        | Male        | 20140628   | 191.0       | 0   |   |
| 1 | Medium Tempo | Pop                | Female      | 20140816   | 368.0       | 0   |   |
| 2 | Medium Tempo | Urban              | Mixed       | 20140118   | 223.0       | 1   |   |
| 3 | Medium Tempo | Alternative & Punk | Male        | 20140104   | 206.0       | 0   |   |
| 4 | Medium Tempo | Traditional        | Female      | 20140301   | 232.0       | 0   |   |

|   | anioNacimiento | edad_en_billboard | moodEncoded |
|---|----------------|-------------------|-------------|
| 0 | 1975.0         | 39                | 4           |
| 1 | 1989.0         | 25                | 6           |
| 2 | 1993.0         | 21                | 5           |
| 3 | 1989.0         | 25                | 4           |
| 4 | 0.0            | 35                | 4           |

```
[20]: # Mapeo del tempo
artists_billboard['tempoEncoded'] = artists_billboard['tempo'].map({'Fast_
↳Tempo': 0,
                                'Medium_
↳Tempo': 2,
                                'Slow_
↳Tempo': 1,
                                ':0}').
↳astype(int)
```

```
[21]: artists_billboard.head()
```

```
[21]:   id      title \
0    0  Small Town Throwdown
1    1      Bang Bang
2    2      Timber
3    3  Sweater Weather
4    4      Automatic
```

|   | artist                                            | mood       | \ |
|---|---------------------------------------------------|------------|---|
| 0 | BRANTLEY GILBERT featuring JUSTIN MOORE & THOM... | Brooding   |   |
| 1 | JESSIE J, ARIANA GRANDE & NICKI MINAJ             | Energizing |   |
| 2 | PITBULL featuring KE\$HA                          | Excited    |   |
| 3 | THE NEIGHBOURHOOD                                 | Brooding   |   |
| 4 | MIRANDA LAMBERT                                   | Yearning   |   |

|   | tempo        | genre       | artist_type | chart_date | durationSeg | top | \ |
|---|--------------|-------------|-------------|------------|-------------|-----|---|
| 0 | Medium Tempo | Traditional | Male        | 20140628   | 191.0       | 0   |   |
| 1 | Medium Tempo | Pop         | Female      | 20140816   | 368.0       | 0   |   |
| 2 | Medium Tempo | Urban       | Mixed       | 20140118   | 223.0       | 1   |   |

|   |              |                    |        |          |       |   |
|---|--------------|--------------------|--------|----------|-------|---|
| 3 | Medium Tempo | Alternative & Punk | Male   | 20140104 | 206.0 | 0 |
| 4 | Medium Tempo | Traditional        | Female | 20140301 | 232.0 | 0 |

|   | anioNacimiento | edad_en_billboard | moodEncoded | tempoEncoded |
|---|----------------|-------------------|-------------|--------------|
| 0 | 1975.0         | 39                | 4           | 2            |
| 1 | 1989.0         | 25                | 6           | 2            |
| 2 | 1993.0         | 21                | 5           | 2            |
| 3 | 1989.0         | 25                | 4           | 2            |
| 4 | 0.0            | 35                | 4           | 2            |

```
[22]: # Mapeo del genero
artists_billboard['genreEncoded'] = artists_billboard['genre'].map(
    {'Urban': 4,
     'Pop': 3,
     'Traditional': 2,
     'Alternative & Punk': 1,
     'Electronica': 1,
     'Rock': 1,
     'Soundtrack': 0,
     'Jazz': 0,
     'Other': 0,
     '': 0}
).astype(int)
```

```
[23]: artists_billboard.head()
```

```
[23]:   id          title \
0    0  Small Town Throwdown
1    1           Bang Bang
2    2           Timber
3    3  Sweater Weather
4    4       Automatic
```

|   | artist                                            | mood \     |
|---|---------------------------------------------------|------------|
| 0 | BRANTLEY GILBERT featuring JUSTIN MOORE & THOM... | Brooding   |
| 1 | JESSIE J, ARIANA GRANDE & NICKI MINAJ             | Energizing |
| 2 | PITBULL featuring KE\$HA                          | Excited    |
| 3 | THE NEIGHBOURHOOD                                 | Brooding   |
| 4 | MIRANDA LAMBERT                                   | Yearning   |

|   | tempo        | genre              | artist_type | chart_date | durationSeg | top \ |
|---|--------------|--------------------|-------------|------------|-------------|-------|
| 0 | Medium Tempo | Traditional        | Male        | 20140628   | 191.0       | 0     |
| 1 | Medium Tempo | Pop                | Female      | 20140816   | 368.0       | 0     |
| 2 | Medium Tempo | Urban              | Mixed       | 20140118   | 223.0       | 1     |
| 3 | Medium Tempo | Alternative & Punk | Male        | 20140104   | 206.0       | 0     |
| 4 | Medium Tempo | Traditional        | Female      | 20140301   | 232.0       | 0     |

|   | anioNacimiento | edad_en_billboard | moodEncoded | tempoEncoded | genreEncoded |
|---|----------------|-------------------|-------------|--------------|--------------|
| 0 | 1975.0         | 39                | 4           | 2            | 2            |
| 1 | 1989.0         | 25                | 6           | 2            | 3            |
| 2 | 1993.0         | 21                | 5           | 2            | 4            |
| 3 | 1989.0         | 25                | 4           | 2            | 1            |
| 4 | 0.0            | 35                | 4           | 2            | 2            |

```
[24]: #Mapeo del tipo de artista
artists_billboard['artist_typeEncoded'] = artists_billboard['artist_type'].map(
    ↪{'Female': 2,
    ↪
    ↪'Male': 3,
    ↪
    ↪'Mixed': 1,
    ↪
    ↪': 0} ).astype(int)
```

```
[25]: artists_billboard.head()
```

```
[25]:   id          title \
0    0  Small Town Throwdown
1    1          Bang Bang
2    2          Timber
3    3  Sweater Weather
4    4        Automatic
```

|   | artist                                            | mood \     |
|---|---------------------------------------------------|------------|
| 0 | BRANTLEY GILBERT featuring JUSTIN MOORE & THOM... | Brooding   |
| 1 | JESSIE J, ARIANA GRANDE & NICKI MINAJ             | Energizing |
| 2 | PITBULL featuring KESHA                           | Excited    |
| 3 | THE NEIGHBOURHOOD                                 | Brooding   |
| 4 | MIRANDA LAMBERT                                   | Yearning   |

|   | tempo        | genre              | artist_type | chart_date | durationSeg | top \ |
|---|--------------|--------------------|-------------|------------|-------------|-------|
| 0 | Medium Tempo | Traditional        | Male        | 20140628   | 191.0       | 0     |
| 1 | Medium Tempo | Pop                | Female      | 20140816   | 368.0       | 0     |
| 2 | Medium Tempo | Urban              | Mixed       | 20140118   | 223.0       | 1     |
| 3 | Medium Tempo | Alternative & Punk | Male        | 20140104   | 206.0       | 0     |
| 4 | Medium Tempo | Traditional        | Female      | 20140301   | 232.0       | 0     |

|   | anioNacimiento | edad_en_billboard | moodEncoded | tempoEncoded | genreEncoded \ |
|---|----------------|-------------------|-------------|--------------|----------------|
| 0 | 1975.0         | 39                | 4           | 2            | 2              |
| 1 | 1989.0         | 25                | 6           | 2            | 3              |
| 2 | 1993.0         | 21                | 5           | 2            | 4              |
| 3 | 1989.0         | 25                | 4           | 2            | 1              |
| 4 | 0.0            | 35                | 4           | 2            | 2              |



|   | artist_typeEncoded |
|---|--------------------|
| 0 | 3                  |
| 1 | 2                  |
| 2 | 1                  |
| 3 | 3                  |
| 4 | 2                  |

```
[26]: #Mapeo de la edad
artists_billboard.loc[ artists_billboard['edad_en_billboard'] <= 21,
↳ 'edadEncoded'] = 0
artists_billboard.loc[(artists_billboard['edad_en_billboard'] > 21) &
↳ (artists_billboard['edad_en_billboard'] <= 26), 'edadEncoded'] = 1
artists_billboard.loc[(artists_billboard['edad_en_billboard'] > 26) &
↳ (artists_billboard['edad_en_billboard'] <= 30), 'edadEncoded'] = 2
artists_billboard.loc[(artists_billboard['edad_en_billboard'] > 30) &
↳ (artists_billboard['edad_en_billboard'] <= 40), 'edadEncoded'] = 3
artists_billboard.loc[ artists_billboard['edad_en_billboard'] > 40,
↳ 'edadEncoded'] = 4
```

```
[27]: artists_billboard.head()
```

```
[27]:
```

|   | id | title \              |
|---|----|----------------------|
| 0 | 0  | Small Town Throwdown |
| 1 | 1  | Bang Bang            |
| 2 | 2  | Timber               |
| 3 | 3  | Sweater Weather      |
| 4 | 4  | Automatic            |

|   | artist                                            | mood \     |
|---|---------------------------------------------------|------------|
| 0 | BRANTLEY GILBERT featuring JUSTIN MOORE & THOM... | Brooding   |
| 1 | JESSIE J, ARIANA GRANDE & NICKI MINAJ             | Energizing |
| 2 | PITBULL featuring KE\$HA                          | Excited    |
| 3 | THE NEIGHBOURHOOD                                 | Brooding   |
| 4 | MIRANDA LAMBERT                                   | Yearning   |

|   | tempo        | genre              | artist_type | chart_date | durationSeg | top \ |
|---|--------------|--------------------|-------------|------------|-------------|-------|
| 0 | Medium Tempo | Traditional        | Male        | 20140628   | 191.0       | 0     |
| 1 | Medium Tempo | Pop                | Female      | 20140816   | 368.0       | 0     |
| 2 | Medium Tempo | Urban              | Mixed       | 20140118   | 223.0       | 1     |
| 3 | Medium Tempo | Alternative & Punk | Male        | 20140104   | 206.0       | 0     |
| 4 | Medium Tempo | Traditional        | Female      | 20140301   | 232.0       | 0     |

|   | anioNacimiento | edad_en_billboard | moodEncoded | tempoEncoded | genreEncoded \ |
|---|----------------|-------------------|-------------|--------------|----------------|
| 0 | 1975.0         | 39                | 4           | 2            | 2              |
| 1 | 1989.0         | 25                | 6           | 2            | 3              |
| 2 | 1993.0         | 21                | 5           | 2            | 4              |
| 3 | 1989.0         | 25                | 4           | 2            | 1              |

|   |     |    |   |   |   |
|---|-----|----|---|---|---|
| 4 | 0.0 | 35 | 4 | 2 | 2 |
|---|-----|----|---|---|---|

|   | artist_typeEncoded | edadEncoded |
|---|--------------------|-------------|
| 0 | 3                  | 3.0         |
| 1 | 2                  | 1.0         |
| 2 | 1                  | 0.0         |
| 3 | 3                  | 1.0         |
| 4 | 2                  | 3.0         |

```
[28]: #Mapeo de duracion en segundos
artists_billboard.loc[ artists_billboard['durationSeg'] <= 150,
↳ 'durationEncoded'] = 0
artists_billboard.loc[(artists_billboard['durationSeg'] > 150) &
↳ (artists_billboard['durationSeg'] <= 180), 'durationEncoded'] = 1
artists_billboard.loc[(artists_billboard['durationSeg'] > 180) &
↳ (artists_billboard['durationSeg'] <= 210), 'durationEncoded'] = 2
artists_billboard.loc[(artists_billboard['durationSeg'] > 210) &
↳ (artists_billboard['durationSeg'] <= 240), 'durationEncoded'] = 3
artists_billboard.loc[(artists_billboard['durationSeg'] > 240) &
↳ (artists_billboard['durationSeg'] <= 270), 'durationEncoded'] = 4
artists_billboard.loc[(artists_billboard['durationSeg'] > 270) &
↳ (artists_billboard['durationSeg'] <= 300), 'durationEncoded'] = 5
artists_billboard.loc[ artists_billboard['durationSeg'] > 300,
↳ 'durationEncoded'] = 6
```

```
[29]: artists_billboard.head()
```

```
[29]:   id          title \
0   0  Small Town Throwdown
1   1           Bang Bang
2   2           Timber
3   3  Sweater Weather
4   4       Automatic
```

|   | artist                                            | mood       | \ |
|---|---------------------------------------------------|------------|---|
| 0 | BRANTLEY GILBERT featuring JUSTIN MOORE & THOM... | Brooding   |   |
| 1 | JESSIE J, ARIANA GRANDE & NICKI MINAJ             | Energizing |   |
| 2 | PITBULL featuring KE\$HA                          | Excited    |   |
| 3 | THE NEIGHBOURHOOD                                 | Brooding   |   |
| 4 | MIRANDA LAMBERT                                   | Yearning   |   |

|   | tempo        | genre              | artist_type | chart_date | durationSeg | top | \ |
|---|--------------|--------------------|-------------|------------|-------------|-----|---|
| 0 | Medium Tempo | Traditional        | Male        | 20140628   | 191.0       | 0   |   |
| 1 | Medium Tempo | Pop                | Female      | 20140816   | 368.0       | 0   |   |
| 2 | Medium Tempo | Urban              | Mixed       | 20140118   | 223.0       | 1   |   |
| 3 | Medium Tempo | Alternative & Punk | Male        | 20140104   | 206.0       | 0   |   |
| 4 | Medium Tempo | Traditional        | Female      | 20140301   | 232.0       | 0   |   |

|   | anioNacimiento | edad_en_billboard | moodEncoded | tempoEncoded | genreEncoded | \ |
|---|----------------|-------------------|-------------|--------------|--------------|---|
| 0 | 1975.0         | 39                | 4           | 2            | 2            |   |
| 1 | 1989.0         | 25                | 6           | 2            | 3            |   |
| 2 | 1993.0         | 21                | 5           | 2            | 4            |   |
| 3 | 1989.0         | 25                | 4           | 2            | 1            |   |
| 4 | 0.0            | 35                | 4           | 2            | 2            |   |

|   | artist_typeEncoded | edadEncoded | durationEncoded |
|---|--------------------|-------------|-----------------|
| 0 | 3                  | 3.0         | 2.0             |
| 1 | 2                  | 1.0         | 6.0             |
| 2 | 1                  | 0.0         | 3.0             |
| 3 | 3                  | 1.0         | 2.0             |
| 4 | 2                  | 3.0         | 3.0             |

```
[30]: # Tirar (drop) todas las columnas que no necesito
drop_elements =
↳ ['id', 'title', 'artist', 'mood', 'tempo', 'genre', 'artist_type', 'chart_date', 'anioNacimiento', '']
```

```
[31]: artists_encoded = artists_billboard.drop(drop_elements, axis = 1)
```

```
[32]: artists_encoded.head()
```

|   | top | moodEncoded | tempoEncoded | genreEncoded | artist_typeEncoded | \ |
|---|-----|-------------|--------------|--------------|--------------------|---|
| 0 | 0   | 4           | 2            | 2            | 3                  |   |
| 1 | 0   | 6           | 2            | 3            | 2                  |   |
| 2 | 1   | 5           | 2            | 4            | 1                  |   |
| 3 | 0   | 4           | 2            | 1            | 3                  |   |
| 4 | 0   | 4           | 2            | 2            | 2                  |   |

|   | edadEncoded | durationEncoded |
|---|-------------|-----------------|
| 0 | 3.0         | 2.0             |
| 1 | 1.0         | 6.0             |
| 2 | 0.0         | 3.0             |
| 3 | 1.0         | 2.0             |
| 4 | 3.0         | 3.0             |

```
[33]: # Entrenar el modelo con todo lo que conlleva el arbol de decisiones
# Encontrar la profundidad máxima de mi árbol
cv = KFold(n_splits=10)
accuracies = []
max_attributes = len(list(artists_encoded))
depth_range = range(1, max_attributes + 1)
```

```
[34]: for depth in depth_range:
    fold_accuracy = []
    tree_model = tree.DecisionTreeClassifier(
```

```

        criterion='entropy',
        min_samples_split=20,
        min_samples_leaf=5,
        max_depth = depth,
        class_weight={1:3.5})
    for train_fold, valid_fold in cv.split(artists_encoded):
        f_train = artists_encoded.loc[train_fold]
        f_valid = artists_encoded.loc[valid_fold]
        model = tree_model.fit(X = f_train.drop(['top'], axis=1), y =
↪f_train["top"])
        valid_acc = model.score(X = f_valid.drop(['top'], axis=1), y =
↪f_valid["top"])
        # calculamos la precision con el segmento de validacion
        fold_accuracy.append(valid_acc)
    avg = sum(fold_accuracy)/len(fold_accuracy)
    accuracies.append(avg)

```

```

[35]: df = pd.DataFrame({"Max Depth": depth_range, "Average Accuracy": accuracies})
df = df[["Max Depth", "Average Accuracy"]]
print(df.to_string(index=False))

```

| Max Depth | Average Accuracy |
|-----------|------------------|
| 1         | 0.556101         |
| 2         | 0.556126         |
| 3         | 0.564038         |
| 4         | 0.653571         |
| 5         | 0.606275         |
| 6         | 0.626736         |
| 7         | 0.640823         |

```

[36]: # Creando el modelo
decision_tree = tree.DecisionTreeClassifier(criterion = "entropy",
                                             min_samples_split = 20,
                                             min_samples_leaf = 5,
                                             max_depth = 4,
                                             class_weight = {1:3.5})

```

```

[37]: # Crear arrays de entrenamiento y las etiquetas que indican si llegó al top o no
y_train = artists_encoded['top']
x_train = artists_encoded.drop(['top'], axis = 1).values

```

```

[38]: decision_tree.fit(x_train, y_train)

```

```

[38]: DecisionTreeClassifier(class_weight={1: 3.5}, criterion='entropy', max_depth=4,
                             min_samples_leaf=5, min_samples_split=20)

```

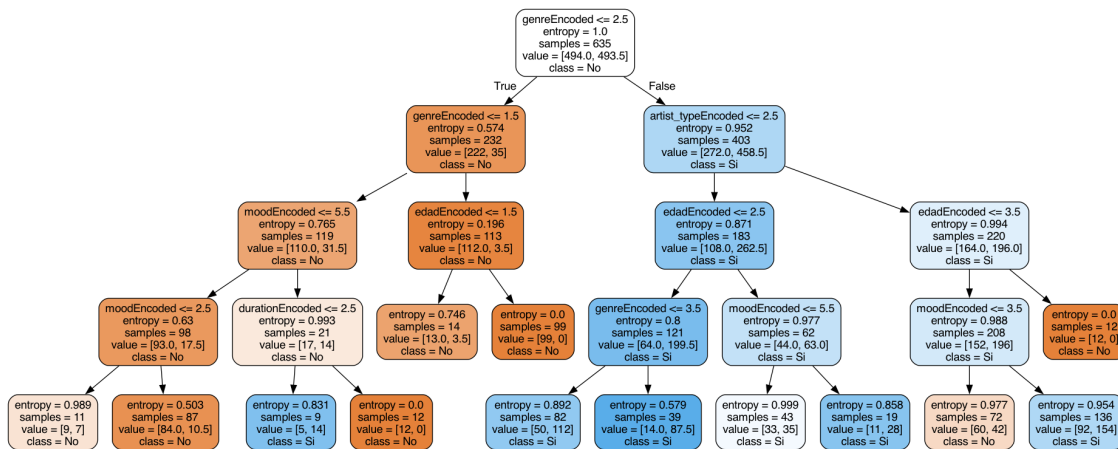
```
[39]: with open(r"tree1.dot", 'w') as f:
      f = tree.export_graphviz(decision_tree,
                              out_file=f,
                              max_depth = 7,
                              impurity = True,
                              feature_names = list(artists_encoded.drop(['top'],
↪axis=1)),

                              class_names = ['No', 'Si'],
                              rounded = True,
                              filled= True )
```

```
[40]: (graph,) = pydot.graph_from_dot_file('tree1.dot')
      graph.write_png('tree1.png')
```

```
[41]: PImage("tree1.png")
```

```
[41]:
```



```
[42]: # Prediccion para la canción Havanna de Camila Cabello que SI llegó al top 1
      x_test = pd.DataFrame(columns=('top', 'moodEncoded', 'tempoEncoded',
↪'genreEncoded', 'artist_typeEncoded', 'edadEncoded', 'durationEncoded '))
      x_test.loc[0] = (1,5,2,4,1,0,3)
```

```
[43]: y_pred = decision_tree.predict(x_test.drop(['top'], axis = 1))
```

```
/Users/haydeml/Library/Python/3.9/lib/python/site-packages/sklearn/base.py:486:
UserWarning: X has feature names, but DecisionTreeClassifier was fitted without
feature names
```

```
warnings.warn(
```

```
[44]: y_pred
```

```
[44]: array([1])
```

```
[45]: y_proba = decision_tree.predict_proba(x_test.drop(['top'], axis = 1))

/Users/haydeml/Library/Python/3.9/lib/python/site-packages/sklearn/base.py:486:
UserWarning: X has feature names, but DecisionTreeClassifier was fitted without
feature names
  warnings.warn(
```

```
[48]: print("Probabilidad de Acierto: " + str(y_proba[0][y_pred]* 100)+"%")
```

Probabilidad de Acierto: [86.20689655]%

```
[49]: # Prediccion para la canción Believer de Imagine Dragons que NO llegó al top 1
x_test = pd.DataFrame(columns=('top', 'moodEncoded', 'tempoEncoded',
    ↳ 'genreEncoded', 'artist_typeEncoded', 'edadEncoded', 'durationEncoded '))
x_test.loc[0] = (0,4,2,1,3,2,3)
```

```
[50]: y_pred = decision_tree.predict(x_test.drop(['top'], axis = 1))

/Users/haydeml/Library/Python/3.9/lib/python/site-packages/sklearn/base.py:486:
UserWarning: X has feature names, but DecisionTreeClassifier was fitted without
feature names
  warnings.warn(
```

```
[51]: y_pred
```

```
[51]: array([0])
```

```
[52]: y_proba = decision_tree.predict_proba(x_test.drop(['top'], axis = 1))
print("Probabilidad de Acierto: " + str(y_proba[0][y_pred]* 100)+"%")
```

Probabilidad de Acierto: [88.88888889]%

```
/Users/haydeml/Library/Python/3.9/lib/python/site-packages/sklearn/base.py:486:
UserWarning: X has feature names, but DecisionTreeClassifier was fitted without
feature names
  warnings.warn(
```

```
[ ]:
```